

화학수학

열 번째 수업

미분방정식

오늘의 수업

- Introduction to differential equations
- First-order ODEs
 - Separable variables
 - Exact equations
 - Linear equations
- Higher-order ODEs
 - Superposition principle
 - Reduction of order



12. Differential Equations

(12장 미분방정식)

Differential Equations (DE, 미분방정식)

Equations that contain derivatives

- Ordinary differential equation (ODE, 상미분방정식) is a differential equation of a single-variable function

$$\frac{d^2y}{dx^2} + \frac{dy}{dx} + x = 5$$

- Partial differential equation (PDE, 편미분방정식) contains partial derivatives

$$\frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} = 0$$

Examples from Physical Chemistry

- Quantum mechanics: 1D Schrödinger equation

$$i\hbar \frac{\partial \Psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} + V\Psi$$

- Kinetics: second-order reaction

$$\frac{d[A]}{dt} = -k[A]^2$$

- Transport phenomena: 1D heat equation

$$\frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2}$$

Terminology

- Order of DE (계수): order of the highest derivative
- Linear (선형) vs. nonlinear (비선형)
 - Linear: contains the function (y) and its derivatives only to the first power (Note: terms in x can be nonlinear)

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

- Nonlinear: otherwise

$$(1 - y) \frac{dy}{dx} + 2y = e^x, \quad \frac{d^2 y}{dx^2} + \sin y = 0, \quad \dots$$



Linear Differential Equations

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

- Homogeneous (동차): $g(x) = 0$
- Inhomogeneous (비동차): $g(x) \neq 0$

General and Particular Solutions

(일반해와 특수해)

- **Solution:** any function that reduces the DE to an identity when substituted into the DE

For a DE of order n ,

- **General solution:** contains n arbitrary constants
- **Particular solution:** contains no arbitrary constants
 - Particular solutions can be generated from the general solution



Example

$$\frac{d^4 y}{dx^4} + 2 \frac{d^2 y}{dx^2} + y = 0$$

- $y = \cos x$ is a particular solution
- $y = -x \sin x$ is another particular solution
- $y = c_1 \cos x + c_2 \sin x + c_3 x \cos x + c_4 x \sin x$ is a general solution

Initial-Value Problems (IVP, 초깃값 문제)

- For a DE of order n , if n additional conditions are given, one can determine the arbitrary constants of the general solution
- Initial conditions (IC, 초기 조건): values of the function and its first $(n - 1)$ derivatives at a single point

$$y(x_0) = y_0, \quad \left. \frac{dy}{dx} \right|_{x=x_0} = y_1, \quad \dots, \quad \left. \frac{d^{n-1}y}{dx^{n-1}} \right|_{x=x_0} = y_{n-1}$$

Example

$$\frac{d^2y}{dx^2} + 4y = 0$$

General solution:

$$y = c_1 \cos 2x + c_2 \sin 2x$$

Initial conditions:

$$y(0) = 1, \quad \left. \frac{dy}{dx} \right|_{x=0} = -6$$

Solution of the IVP:

$$y = \cos 2x - 3 \sin 2x$$

Boundary-Value Problems (BVP, 경계값 문제)

- For a DE of order n , if n additional conditions are given, one can determine the arbitrary constants of the general solution
- Boundary conditions (BC, 경계 조건): values of the function and its derivatives at different points

Example

$$\frac{d^2y}{dx^2} + 4y = 0$$

General solution:

$$y = c_1 \cos 2x + c_2 \sin 2x$$

Boundary conditions:

$$y(0) = 1, \quad y(\pi/4) = -2$$

Solution of the BVP:

$$y = \cos 2x - 2 \sin 2x$$

Solving a Differential Equation

1. Obtain the **general solution**: hardest part!
 2. If initial/boundary conditions are given, substitute the conditions to **determine the coefficients**
- No universal method to solve all DEs
 - For specific forms of DEs, we have general solving methods

First-Order ODEs

$$F\left(y, \frac{dy}{dx}\right) = 0$$

- Separable variables (§ 12.4)
- Exact equations (§ 12.5, 12.6)
- Linear equations (not in Mortimer)

Separable Variables

- Separable DE:

$$g(y) \frac{dy}{dx} = f(x)$$

This DE can be manipulated into

$$g(y) dy = f(x) dx$$

$$\int g(y) dy = \int f(x) dx + C$$

This equation provides the **general solution**



Example 12.9

In a first-order chemical reaction with no back reaction, the concentration of the reactant is governed by

$$-\frac{dc}{dt} = kc$$

where c is the concentration, t is the time, and k is the rate constant. Solve the equation to find c as a function of t .

Rearrange the equation to separate variables:

$$\frac{1}{c} dc = -k dt$$

$$\int \frac{1}{c} dc = - \int k dt$$

Example 12.9

$$-\frac{dc}{dt} = kc$$

Solve the equation to find c as a function of t .

$$\int \frac{1}{c} dc = - \int k dt$$

Integration leads to

$$\ln c = -kt + C$$

$$c = e^{-kt+C} = Ae^{-kt}$$

This is the general solution and A is an arbitrary constant.

Exact Differential Equations

- Pfaffian form:

$$M(x, y) dx + N(x, y) dy = 0$$

If the differential equation is **exact**,
there is a function $f(x, y)$ such that

$$df = M(x, y) dx + N(x, y) dy = 0$$

which implies that $f(x, y) = k$ (constant)

How to Find $f(x, y)$

$$df = M(x, y) dx + N(x, y) dy = 0$$

$$f(x_1, y_1) = f(x_0, y_0) + \int_C df$$

where C is a curve connecting (x_0, y_0) and (x_1, y_1) . Take a rectangular path from (x_0, y_0) to (x_1, y_0) to (x_1, y_1) :

$$f(x_1, y_1) = f(x_0, y_0) + \int_{x_0}^{x_1} M(x, y_0) dx + \int_{y_0}^{y_1} N(x_1, y) dy$$

This gives the relationship between x and y .

Example 12.10

Solve the differential equation

$$2xy \, dx + x^2 dy = 0.$$

Check the exactness first:

$$\frac{\partial(2xy)}{\partial y} = 2x, \quad \frac{\partial(x^2)}{\partial x} = 2x$$

Do a line integral from (x_0, y_0) to (x_1, y_0) to (x_1, y_1) :

$$f(x_1, y_1) = f(x_0, y_0) + \int_{x_0}^{x_1} M(x, y_0) \, dx + \int_{y_0}^{y_1} N(x_1, y) \, dy$$

$$f(x_1, y_1) = f(x_0, y_0) + \int_{x_0}^{x_1} (2xy_0) \, dx + \int_{y_0}^{y_1} (x_1^2) \, dy$$

Example 12.10

Solve the differential equation

$$2xy \, dx + x^2 dy = 0.$$

$$f(x_1, y_1) = f(x_0, y_0) + \int_{x_0}^{x_1} (2xy_0) \, dx + \int_{y_0}^{y_1} (x_1^2) \, dy$$

$$f(x_1, y_1) = f(x_0, y_0) + (x_1^2 y_0 - x_0^2 y_0) + (x_1^2 y_1 - x_1^2 y_0)$$

Hence,

$$f(x_1, y_1) - f(x_0, y_0) = x_1^2 y_1 - x_0^2 y_0$$

which implies that $f(x, y) = x^2 y$.

Example 12.10

Solve the differential equation

$$2xy \, dx + x^2 dy = 0.$$

$$f(x, y) = x^2 y$$

Now, we impose $f(x, y) = k$:

$$x^2 y = k$$

Therefore,

$$y = \frac{k}{x^2}$$

This is the general solution and k is an arbitrary constant.

Integrating Factors

- Pfaffian form:

$$M(x, y) dx + N(x, y) dy = 0$$

If the differential equation is **inexact**, but we know the **integration factor** $g(x, y)$ that makes

$$g(x, y)M(x, y) dx + g(x, y)N(x, y) dy = 0$$

an exact differential equation, we can solve this DE.

Example 12.11

Solve the differential equation

$$y \, dx - x \, dy = 0$$

where $1/x^2$ is an integrating factor.

Check the exactness first:

$$\frac{\partial(y)}{\partial y} = 1, \quad \frac{\partial(-x)}{\partial x} = -1$$

Hence, the differential equation is inexact.

Multiplying the integrating factor, we obtain

$$\frac{y}{x^2} \, dx - \frac{1}{x} \, dy = 0$$

Example 12.11

Solve the differential equation

$$y \, dx - x \, dy = 0$$

where $1/x^2$ is an integrating factor.

$$\frac{y}{x^2} \, dx - \frac{1}{x} \, dy = 0$$

Check the exactness:

$$\frac{\partial}{\partial y} \left(\frac{y}{x^2} \right) = \frac{1}{x^2}, \quad \frac{\partial}{\partial x} \left(-\frac{1}{x} \right) = \frac{1}{x^2}$$

Now the differential equation is exact.

Example 12.11

Solve the differential equation

$$y \, dx - x \, dy = 0$$

where $1/x^2$ is an integrating factor.

$$\frac{y}{x^2} \, dx - \frac{1}{x} \, dy = 0$$

Do a line integral from (x_0, y_0) to (x_1, y_0) to (x_1, y_1) :

$$f(x_1, y_1) = f(x_0, y_0) + \int_{x_0}^{x_1} M(x, y_0) \, dx + \int_{y_0}^{y_1} N(x_1, y) \, dy$$

$$f(x_1, y_1) = f(x_0, y_0) + \int_{x_0}^{x_1} \left(\frac{y_0}{x^2} \right) \, dx + \int_{y_0}^{y_1} \left(-\frac{1}{x_1} \right) \, dy$$

Example 12.11

Solve the differential equation

$$y \, dx - x \, dy = 0$$

where $1/x^2$ is an integrating factor.

$$f(x_1, y_1) = f(x_0, y_0) + \int_{x_0}^{x_1} \left(\frac{y_0}{x^2} \right) dx + \int_{y_0}^{y_1} \left(-\frac{1}{x_1} \right) dy$$

$$f(x_1, y_1) = f(x_0, y_0) + y_0 \left(-\frac{1}{x_1} + \frac{1}{x_0} \right) - \frac{1}{x_1} (y_1 - y_0)$$

$$f(x_1, y_1) - f(x_0, y_0) = \frac{y_0}{x_0} - \frac{y_1}{x_1}$$

This implies that $f(x, y) = -y/x$.

Example 12.11

Solve the differential equation

$$y \, dx - x \, dy = 0$$

where $1/x^2$ is an integrating factor.

$$f(x, y) = -y/x$$

Now, we impose $f(x, y) = k$:

$$-y/x = k$$

Therefore,

$$y = k'x$$

This is the general solution and k' is an arbitrary constant.

Linear Equations

$$a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

- Standard form:

$$\frac{dy}{dx} + P(x)y = f(x)$$

where $P(x) = a_0(x)/a_1(x)$ and $f(x) = g(x)/a_1(x)$

Solving the Standard Form

$$\frac{dy}{dx} + P(x)y = f(x)$$

Multiply both sides by $e^{\int P(x)dx}$;

$$e^{\int P(x)dx} \frac{dy}{dx} + e^{\int P(x)dx} P(x)y = e^{\int P(x)dx} f(x)$$

$$\frac{d}{dx} [e^{\int P(x)dx} y] = e^{\int P(x)dx} f(x)$$

Solving the Standard Form

$$\frac{d}{dx} [e^{\int P(x) dx} y] = e^{\int P(x) dx} f(x)$$

Integration leads to

$$e^{\int P(x) dx} y = \int e^{\int P(x) dx} f(x) dx + C$$

Hence,

$$y = e^{-\int P(x) dx} \int e^{\int P(x) dx} f(x) dx + C e^{-\int P(x) dx}$$

Example

Solve the differential equation

$$3 \frac{dy}{dx} + 12y = 4.$$

The standard form is

$$\frac{dy}{dx} + 4y = \frac{4}{3}$$

and $P(x) = 4$, which gives $e^{\int P(x)dx} = e^{\int 4dx} = e^{4x}$.

$$e^{4x} \frac{dy}{dx} + 4e^{4x}y = \frac{4}{3}e^{4x}$$

$$\frac{d}{dx}(e^{4x}y) = \frac{4}{3}e^{4x}$$

Example

Solve the differential equation

$$3 \frac{dy}{dx} + 12y = 4.$$

$$\frac{d}{dx}(e^{4x}y) = \frac{4}{3}e^{4x}$$

Integration leads to

$$e^{4x}y = \frac{4}{3} \int e^{4x} dx + C = \frac{1}{3}e^{4x} + C$$

Hence,

$$y = \frac{1}{3} + Ce^{-4x}$$

Another Example

Solve the differential equation

$$\frac{dy}{dx} + y = x$$

subject to $y(0) = 4$.

The equation is in standard form and

$P(x) = 1$, which gives $e^{\int P(x)dx} = e^{\int 1dx} = e^x$.

$$e^x \frac{dy}{dx} + e^x y = xe^x$$

$$\frac{d}{dx}(e^x y) = xe^x$$

Another Example

Solve the differential equation

$$\frac{dy}{dx} + y = x$$

subject to $y(0) = 4$.

$$\frac{d}{dx}(e^x y) = x e^x$$

Integration leads to

$$e^x y = \int x e^x dx + C = x e^x - e^x + C$$

Hence,

$$y = x - 1 + C e^{-x}$$

Another Example

Solve the differential equation

$$\frac{dy}{dx} + y = x$$

subject to $y(0) = 4$.

$$y = x - 1 + Ce^{-x}$$

The initial condition states that for $x = 0$, $y = 4$. Substitution gives

$$4 = 0 - 1 + C$$

Thus, $C = 5$ and the solution is

$$y = x - 1 + 5e^{-x}$$

Higher-Order ODEs: Linear Case

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

- Superposition principle (§ 12.2.1)
- Reduction of order (not in Mortimer)
- Linear ODEs with constant coefficients (§ 12.2, 12.3)
 - Homogeneous (§ 12.2)
 - Inhomogeneous (§ 12.3)
- We will not discuss nonlinear ODEs

Homogeneous Linear Equations

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = 0$$

- Superposition principle (중첩 원리):

If $y_1(x), y_2(x), \dots, y_k(x)$ are the solutions of the ODE, the linear combination

$$y = c_1 y_1(x) + c_2 y_2(x) + \cdots + c_k y_k(x)$$

is also a solution of the ODE for arbitrary constants $c_1, c_2, \dots, c_k$.

Example

$y_1 = \cos x$ and $y_2 = \sin x$ are the solutions of the differential equation

$$\frac{d^2y}{dx^2} + y = 0.$$

Show that a linear combination of y_1 and y_2 is also its solution.

For arbitrary constants c_1 and c_2 , the linear combination is

$$y = c_1 y_1 + c_2 y_2 = c_1 \cos x + c_2 \sin x$$

Substitution leads to

$$\begin{aligned} \frac{d^2y}{dx^2} + y &= \frac{d^2}{dx^2} (c_1 \cos x + c_2 \sin x) + (c_1 \cos x + c_2 \sin x) \\ &= -c_1 \cos x - c_2 \sin x + (c_1 \cos x + c_2 \sin x) = 0 \end{aligned}$$

Homogeneous Linear Second-Order ODEs

$$a_2(x) \frac{d^2 y}{dx^2} + a_1(x) \frac{dy}{dx} + a_0(x)y = 0$$

If a solution y_1 is known,
we can obtain **the other solution y_2**
by following a systematic procedure
→ **reduction of order** (계수내림)

Reduction of Order

- Standard form

$$\frac{d^2y}{dx^2} + P(x) \frac{dy}{dx} + Q(x)y = 0$$

- With known $y_1(x)$, define $y(x) = u(x)y_1(x)$:

$$\frac{dy}{dx} = u \frac{dy_1}{dx} + y_1 \frac{du}{dx}$$

$$\frac{d^2y}{dx^2} = u \frac{d^2y_1}{dx^2} + 2 \frac{du}{dx} \frac{dy_1}{dx} + y_1 \frac{d^2u}{dx^2}$$

Substitution!

Rearrangement

$$\frac{d^2 y}{dx^2} + P(x) \frac{dy}{dx} + Q(x)y$$

$$= u \left(\frac{d^2 y_1}{dx^2} + P \frac{dy_1}{dx} + Q y_1 \right) + y_1 \frac{d^2 u}{dx^2} + \left(2 \frac{dy_1}{dx} + P y_1 \right) \frac{du}{dx}$$

$= 0$

$$= y_1 \frac{d^2 u}{dx^2} + \left(2 \frac{dy_1}{dx} + P y_1 \right) \frac{du}{dx}$$

$$= 0$$

“Reduction of Order”

$$y_1 \frac{d^2 u}{dx^2} + \left(2 \frac{dy_1}{dx} + P y_1 \right) \frac{du}{dx} = 0$$

Let $w = du/dx$;

$$y_1 \frac{dw}{dx} + \left(2 \frac{dy_1}{dx} + P y_1 \right) w = 0$$

(First-order DE)

Solving First-Order ODE

$$y_1 \frac{dw}{dx} + \left(2 \frac{dy_1}{dx} + P y_1 \right) w = 0$$

Separation of variables:

$$\frac{dw}{w} = - \frac{2(dy_1/dx)}{y_1} dx - P dx$$

Integration:

$$\ln w = - \ln y_1^2 - \int P dx + C$$

$$w y_1^2 = C' e^{-\int P dx}$$

Going Back to $u(x)$

$$w y_1^2 = C' e^{-\int P dx}$$

$$w = \frac{C'}{y_1^2} e^{-\int P dx}$$

Since $w = du/dx$,

$$\frac{du}{dx} = \frac{C'}{y_1^2} e^{-\int P dx}$$

$$u = \int \frac{C'}{y_1^2} e^{-\int P dx} dx + C''$$

Another Solution

$$u = \int \frac{C'}{y_1^2} e^{-\int P dx} dx + C''$$

For $y(x) = u(x)y_1(x)$, choose $C' = 1$ and $C'' = 0$:

$$y_2(x) = y_1 \int \frac{1}{y_1^2} e^{-\int P dx} dx$$

Example

The function $y_1 = x^2$ is a solution of

$$x^2 \frac{d^2 y}{dx^2} - 3x \frac{dy}{dx} + 4y = 0.$$

Find the general solution.

The standard form of the equation is

$$\frac{d^2 y}{dx^2} - \frac{3}{x} \frac{dy}{dx} + \frac{4}{x^2} y = 0$$

and $P(x) = -3/x$. Now use

$$y_2(x) = y_1 \int \frac{1}{y_1^2} e^{-\int P dx} dx$$

Example

The function $y_1 = x^2$ is a solution of

$$x^2 \frac{d^2 y}{dx^2} - 3x \frac{dy}{dx} + 4y = 0.$$

Find the general solution.

$$y_2(x) = y_1 \int \frac{1}{y_1^2} e^{-\int P dx} dx, \quad P(x) = -\frac{3}{x}$$

Substitution leads to

$$y_2(x) = x^2 \int \frac{1}{x^4} e^{\int \frac{3}{x} dx} dx = x^2 \int \frac{1}{x^4} e^{3 \ln x} dx = x^2 \int \frac{1}{x^4} x^3 dx = x^2 \int \frac{1}{x} dx$$

$$y_2(x) = x^2 \ln x$$

Example

The function $y_1 = x^2$ is a solution of

$$x^2 \frac{d^2 y}{dx^2} - 3x \frac{dy}{dx} + 4y = 0.$$

Find the general solution.

$$y_2(x) = x^2 \ln x$$

The general solution is a linear combination of $y_1(x)$ and $y_2(x)$:

$$y(x) = c_1 y_1(x) + c_2 y_2(x) = c_1 x^2 + c_2 x^2 \ln x$$